Geographic Epistemic Asymmetries in LLM-Assisted Policy Research: A 15-Country, 6-Model, 4-Language Benchmark

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Abstract

Public environmental managers increasingly delegate air pollution policy questions to large language models (LLMs), yet whether these systems answer as reliably for the Global South as for the Global North, and whether prompt framing changes the answer, remains unmeasured for this high-stakes domain. $PM_{2.5}$ is among the leading environmental risk factors for premature mortality, and the burden falls disproportionately on the Global South, where corpus and linguistic under-representation plausibly shape model knowledge [1–3]. We present a *15-country, 6-model, 4-language* benchmark in a single applied domain, air pollution policy, anchored to official ground truth across five task types (normative recall, local measured data, health-evidence synthesis, policy instruments, and applied recommendation), with a pre-specified persona factor (neutral versus public environmental manager) tested within subject and a within-country English/native-language contrast (Spanish, Portuguese, Hindi). From LLM-as-judge composite scores we find: (i) a small but consistent geographic gradient — the Global North outscores the Global South by 6.5 percentage points (Cohen's $d = +0.25$) (H1); (ii) native-language prompting *lowers* accuracy (-3.5 pp overall, -11.6 pp for Hindi versus -0.5 to -1.8 pp for Latin-script languages), so asking in the local language does not help and can hurt (H2); (iii) persona prompting does *not* narrow the gap (difference-in-differences $+0.6$ pp), so prompt framing is not a fix for geographic bias (H6 not supported), consistent with evidence that expert personas do not improve factual accuracy [4, 5]; (iv) the Brazilian-Portuguese regional model is the weakest of all models, contradicting the optimistic regional-model hypothesis (H3); and (v) failure concentrates in technical-standard and local-datum recall (≈ 0.32), a regulatory-cutoff failure mode. Two intuitive remedies — a local-manager persona and the local language — both fail. Inter-judge reliability against an out-of-sample judge is strong (ICC $= 0.82$), and the technical-standard ground truth is anchored to official primary sources for all 15 countries. The study uses a single (reliability-checked) judge rather than the planned multi-judge ensemble, and defers the corpus-representation mechanism (H4) to the final analysis. Protocol, prompts, and

analysis code will be released on publication.

Keywords: large language models; geographic bias; Global South; epistemic justice; policy research; benchmark; multilingual NLP

1 Introduction

Large language models (LLMs) have moved from research curiosity to working infrastructure of public administration in under five years. Environmental agencies, municipal secretariats, and applied policy analysts increasingly draft technical syntheses, translate regulatory norms, locate official monitoring data, and characterize institutional frameworks with LLM assistance [6, 7]. Air pollution policy is a domain where this adoption carries direct stakes for public health. Fine particulate matter ($PM_{2.5}$) is among the leading environmental risk factors for premature mortality worldwide, and the burden falls disproportionately on the Global South [1]. When a public environmental manager in São Paulo, Lagos, or Jakarta asks an LLM about an ambient air quality standard, an attainment deadline, or the epidemiological evidence behind an intervention, an inaccurate answer is not a benign error. It can propagate into a regulatory brief, a procurement decision, or an emergency response during a pollution episode.

The question this paper addresses is whether LLMs serve Global South air pollution policy as reliably as they serve the Global North, and whether the way a query is framed changes the answer. Two recent strands of evidence make this question urgent rather than hypothetical.

Geographic factual bias is real and measurable. Moayeri et al. [8] documented 1.5× higher factual error rates for Sub-Saharan African countries versus North American countries across 20 models and 11 World Bank indicators. Manvi et al. [9] showed that LLM zero-shot ratings on subjective topics correlate strongly with income level (ρ up to 0.70 against socioeconomic conditions). Mirza et al. [10] found that GPT-4 systematically privileges statements about the Global North over the Global South in factual verification tasks. These results establish the phenomenon, but none of them targets a single high-stakes applied domain with verifiable official ground truth, and none asks whether the bias can be modulated by prompt framing.

Linguistic and corpus asymmetry is structural. Joshi et al. [2] mapped a six-class hierarchy of linguistic resource availability in NLP, showing that the languages of most Global South countries sit in the under-served lower classes. Under-representation in the training corpus is the dominant explanation for downstream geographic bias [11]. This asymmetry is not incidental; it reproduces longstanding patterns of uneven digital and epistemic infrastructure that decolonial scholarship has analysed for AI systems [3].

We identify three gaps that current work leaves open for a domain like air pollution policy.

Gap 1: no benchmark targets air pollution policy with official ground truth. Existing evaluations use numeric World Bank recall (WorldBench), subjective ratings (Manvi et al.), journalistic fact-checking (Global-Liar), or everyday cultural knowledge (BLEnD [12]). None measures the recall and synthesis tasks a public environmental manager actually delegates to an

LLM: stating a binding ambient standard, retrieving an officially measured concentration, summarizing the health-burden evidence, identifying the legal instruments and executing agencies, or recommending an operational response. These tasks anchor to authoritative sources (national environmental councils, state monitoring agencies, the World Health Organization, the Global Burden of Disease estimates), which makes accuracy verifiable in a way that trivia benchmarks are not.

Gap 2: persona framing is unexamined as an experimental lever. Practitioners routinely prefix queries with a role ("As a municipal environmental secretary, ..."), and persona or role conditioning is widely used in deployed prompting workflows. Whether adopting the persona of a public environmental manager *reduces* the Global South accuracy gap (by steering the model toward policy-relevant, regulation-compliant responses) or *amplifies* it (by inviting confident fabrication of regulatory specifics the model never learned) has not been tested in a pre-specified design. We treat persona as a within-subject experimental factor rather than an uncontrolled confound.

Gap 3: the corpus-representation mechanism is asserted, not tested for this domain. The link between corpus representation and accuracy is assumed but rarely measured against confounders. We test whether country-level corpus-representation proxies (Wikipedia article counts as primary, Common Crawl shares as secondary) correlate with air pollution policy accuracy after adjusting for Human Development Index and GDP per capita.

Contributions of this work

We present a study protocol for a benchmark designed for these gaps, with hypotheses, sampling, and analysis plan pre-specified before confirmatory data collection [13]. Our contributions are:

1. **An air pollution policy task suite with verifiable ground truth.** We construct five task types mapped to real decision activities of a public environmental manager: recall of binding ambient standards and emission limits (T1), retrieval of officially measured local concentrations (T2), synthesis of epidemiological health-burden evidence (T3), identification of policy instruments and executing agencies (T4), and rubric-scored applied recommendation for an operational scenario (T5). Each task anchors to official national or international sources, and every reference in the ground truth is verified against a resolvable DOI or URL before entering the dataset.
2. **Persona prompting as a pre-specified experimental factor.** We manipulate persona within subject (neutral query vs. public environmental manager) and test, as a co-primary confirmatory hypothesis (H6), whether the persona condition reduces the Global South accuracy gap. We pre-specify both the reduction direction (Norm-Compliance hypothesis) and the amplification direction (Persona-Vacuum hypothesis), reporting whichever the data support.
3. **Theoretically-triangulated country sampling.** We select 15 countries stratified along independent axes: UNCTAD North/South classification (operationalising coloniality [3]),

the Joshi et al. six-class linguistic-resource taxonomy [2], and Human Development and income indicators. This separates geographic, linguistic, and economic contributions to the bias.

4. **Mechanistic test with sensitivity analysis.** We test the corpus-representation hypothesis (H4) using country-level Wikipedia and Common Crawl proxies, with partial-correlation controls for HDI and log GDP per capita and an E-value sensitivity threshold [14].

5. **Regional-model probe.** We include a Brazilian-Portuguese instruction-tuned open-weight model (Cabra-Mistral 7B v3) to test, as a secondary confirmatory hypothesis (H3), whether regional tuning narrows the Portuguese-language gap for Brazil on air pollution policy tasks.

6. **Open resources.** The full protocol, prompts (after validation), ground-truth source documents, analysis code, and figures will be released on publication under CC-BY-4.0 and MIT licenses.

Theoretical framing

We interpret our findings within the *coloniality of knowledge* framework [3, 15] and its data and infrastructural articulations [16, 17]. LLMs are compressed statistical summaries of corpora produced under conditions of uneven digital infrastructure and uneven publication access. The languages and regions that Joshi et al. place in the under-served classes are precisely those where air pollution kills the most people [1, 2]. When a Global South environmental manager consults a model trained largely on Global North text to govern a local air quality problem, the resulting decision risks inheriting the asymmetries of the corpus. This positions our empirical work as a test of one link in a theorised reproduction cycle (training-corpus asymmetry → LLM knowledge asymmetry → policy decision asymmetry), and the persona manipulation as a test of whether prompt framing can interrupt that link.

The paper proceeds as follows. Section 2 positions our work relative to the literature. Section 3 presents the pre-specified design. Section 4 reports the empirical findings. Section 5 interprets them through both statistical and critical lenses and considers practical implications for air pollution governance. Section 6 closes.

2 Related Work

We organize prior work along four threads: the discovery and measurement of geographic factual bias, multilingual and cultural extensions, alignment and mitigation attempts, and LLM use in policy and health decision contexts. For each we state how the air pollution policy focus and the persona manipulation of the present study extend the thread.

2.1 Geographic factual bias

Manvi et al. [9] established that frontier LLMs carry systematic biases against regions of lower socioeconomic conditions, with Spearman correlations up to 0.70 between model-generated ratings of attractiveness, morality, and intelligence and per-capita income. The outcome space (subjective ratings) leaves open whether the same bias appears in objective recall of regulatory and epidemiological facts, which is what air pollution policy requires.

Moayeri et al. [8] introduced WorldBench, quantifying factual recall against World Bank indicators across 20 LLMs and 11 indicators and documenting 1.5× higher absolute relative error for Sub-Saharan Africa versus North America. WorldBench also pioneered automatic detection of citation hallucination, where models cite the World Bank while producing false statistics. The design is bounded by the macro-economic indicator space; an environmental manager asks narrower, source-specific questions (a national ambient standard, an attainment phase deadline, a city-year measured concentration) whose ground truth lives in environmental councils and monitoring agencies rather than in a single aggregated database.

Mirza et al. [10] (Global-Liar) extended the question to temporal stability, showing that newer GPT-4 versions do not monotonically improve and that the privileging of Global North statements persists across versions. Their contribution is temporal and within-model rather than cross-model, cross-language, or domain-specific.

These studies establish that the phenomenon exists and is measurable. None isolates a single applied domain with verifiable official ground truth, and none treats prompt framing as a manipulable factor.

2.2 Multilingual and cultural extensions

Myung et al. [12] (BLEnD) introduced question-answer pairs covering 16 countries and 13 languages in everyday cultural knowledge. Their finding that LLMs perform better in the native languages of high-resource cultures but worse in those of low-resource ones motivates our exploratory language hypothesis (H2) and connects directly to the resource hierarchy of Joshi et al. [2]. Yet BLEnD measures cultural-knowledge accuracy, not the accuracy of regulatory and health-policy recall, and its country selection is not stratified theoretically.

Regional benchmarks proliferated: BRoverbs [18] for Brazilian proverbs, TiEBe [19] for regional historical events, and ALBA [20] and AMALIA [21] for European Portuguese. These document concrete failures in culturally grounded tasks but do not address applied policy use, and each is limited to one or two languages. Work on Portuguese corpus construction [22] underscores how Common Crawl snapshots under-represent Lusophone content, the corpus asymmetry our H4 tests.

2.3 Alignment and mitigation

Opuszko and Böhm [23] investigated whether reasoning models reduce geographic bias, finding that reasoning helps in aggregate but does not eliminate regional disparities. Kerche et al. [24] proposed a place-based typology of LLM biases, situating geographic bias within a broader sociotechnical landscape. These contributions are conceptual and aggregate; we add a domain-specific, pre-specified empirical test and introduce persona as a candidate mitigation lever rather than a fixed property of the model.

2.4 LLMs in policy and health decision contexts

A parallel literature examines LLM use in specific policy settings. He et al. [25] evaluated LLM policy recommendations for homelessness across four cities (including Barcelona, Johannesburg, and South Bend), finding "context-blind rigidity" in which models apply stable internal heuristics poorly calibrated to local realities and align more closely with Global North experts. Their result shows the bias matters for policy in a single domain and a single task type (recommendation). We generalize to a regulatory and health domain with five task types spanning recall, synthesis, and recommendation, and we add the persona manipulation absent from their design.

In health specifically, Kozlakidis et al. [26] argue that LLM hallucinations carry acute risk in regulatory environments with limited technical capacity, noting that language and cultural mismatch between globally trained models and local practice increases error. Air pollution policy sits at the intersection of this risk and the mortality burden documented by the Global Burden of Disease estimates [1], which makes accurate retrieval of standards and health evidence a public-health concern rather than a benchmarking abstraction.

2.5 Persona and role conditioning

Role or persona conditioning is a common practice in deployed prompting, and a growing body of work reports that assigning an expert role can shift model outputs in either direction depending on task and domain. Systematic evidence is sobering: across four LLM families and 2,410 factual questions, adding expert personas to system prompts did not improve accuracy over a no-persona control, and across 162 roles it slightly *reduced* accuracy on average [4]; an independent replication likewise finds that expert personas do not improve factual accuracy [5]. Whether persona conditioning helps or harms factual reliability in high-stakes, knowledge-intensive domains is therefore unsettled and, to our knowledge, untested for geographic equity. We therefore treat persona as a pre-specified experimental factor and test both the Norm-Compliance hypothesis (persona reduces the gap) and the Persona-Vacuum hypothesis (persona amplifies it).

2.6 Positioning of the present work

Our design differs from prior work in four concrete ways: (i) a single high-stakes applied domain (air pollution policy) with verifiable official ground truth, rather than numeric trivia or cultural knowledge; (ii) persona as a pre-specified within-subject experimental factor, rather than an uncontrolled framing choice; (iii) theory-driven country sampling on independent axes (UNCTAD coloniality, Joshi linguistic resource, development), rather than regional convenience; and (iv) an explicit corpus-representation mechanism test with sensitivity analysis. No prior study combines a regulated public-health domain, a persona manipulation, and a pre-specified mechanism test.

3 Methods

3.1 Research design

We specify a pre-specified, factorial, multi-country benchmark experiment to quantify geographic and persona-modulated bias in 14 large language models on air pollution policy tasks relevant to public environmental management. The design crosses 15 countries (stratified along three theoretically grounded axes), 14 LLMs spanning five access tiers, one theory-driven domain (air pollution policy), 5 task types, and 2 persona conditions (neutral versus public environmental manager), with 2 stochastic replications per prompt-model-persona cell. The core stimulus set comprises approximately 600 unique prompts (40 per country across the five tasks and a sparse multilingual matrix), targeting 33,600 model responses ($600 \times 14 \times 2$ personas$\times 2$ reps). The study protocol fixes, in advance of confirmatory data collection, the hypotheses, primary models, persona protocol, exclusion criteria, and multiple-comparison corrections. A Brazil-only collection executed under an earlier three-domain design is reclassified as an extended pilot and is reported separately; it informed prompt calibration but contributes no confirmatory observations.

3.2 Country selection

Countries were selected via theoretical sampling [27] triangulated across three independent classifications. First, the **UNCTAD Global North/South classification** [28] operationalizes the coloniality-of-knowledge framework that motivates this study [3]. Second, the **Joshi et al. (2020) six-class taxonomy** of linguistic-resource representation in NLP corpora provided an orthogonal stratification dimension [2]. Third, **World Bank income groups (FY26)** [29] captured economic position independently of the previous two axes.

The final sample of 15 countries — Brazil, Mexico, Argentina, Peru (Latin America); Nigeria, South Africa, Kenya, Egypt (Africa); India, Indonesia, Bangladesh, the Philippines (Asia); and the United States, Germany, Japan (Global North reference) — covers four world regions, five Joshi classes (1 through 5), and four World Bank income groups. The twelve Global South countries also vary in domestic air quality governance maturity, which is relevant to ground-truth availability (Section 3.8). Countries with compromised statistical governance (e.g., active conflict) or without a publicly retrievable national ambient air quality standard were excluded; Oceania and Joshi Class 0 languages were not represented given feasibility constraints and are acknowledged as scope limitations.

3.3 Country covariates

For each country we compiled five covariates used in the gradient and mechanism tests. **Human Development Index (HDI)** and **GDP per capita** index developmental position. **Joshi class** indexes linguistic representation in NLP corpora [2]. Two corpus-representation proxies index training-data presence: **Wikipedia article and pageview counts** per relevant language (primary proxy for H4) and **Common Crawl** token volume per country, the latter operationalized three ways (country-code top-level domain aggregation, language share, and language $\times$ country intersection) to support a convergent-validity criterion. HDI and GDP enter the mechanism analysis as adjustment covariates; the corpus proxies are the exposures of interest. All covariate values, sources, and access dates are versioned in the protocol release.

3.4 Model selection

Fourteen LLMs were benchmarked across five access tiers, spanning roughly two orders of magnitude in parameter count and four training-data geographies (United States, China, Europe, Brazil). **Tier A (open-weight frontier, 5 models)** comprised large open-weight systems accessed via free inference endpoints. **Tier B (open-weight mid, 3 models)** and **Tier C (open-weight small/regional, 2 models)** were run locally via Ollama under pinned model tags; Tier C includes a Brazilian-Portuguese instruction-tuned open model (Cabra-Mistral 7B v3) used for the regional-model test (H3) against a scale-matched globally trained open model. **Tier D (closed accessible, 2 models)** and **Tier E (closed frontier, 1 model)** were accessed via official APIs. Each model was queried with an identical version string or pinned tag throughout the collection window; tags and, where available, weight hashes are recorded in the protocol release to guard against mid-collection model updates. The complete model roster with vendor, parameter count, access route, and execution backend is provided in Supplementary Table S1.

3.5 Domain and task taxonomy

The single domain is **air pollution policy**, chosen because it anchors to officially published, verifiable ground truth (national ambient air quality standards, environmental-agency monitoring data, federal control programs) and maps to concrete decision activities of a public environmental manager. Five task types operationalize the domain:

- **T1 — Technical standard.** Recall of binding ambient air quality standards, emission limits, or regulatory thresholds (for example, the annual mean $PM_{2.5}$ standard in force in the country).
- **T2 — Local factual datum.** Recall of officially measured air quality data for a specified city or region and year, as reported by the national or subnational environmental agency.
- **T3 — Health-evidence synthesis.** Synthesis of peer-reviewed epidemiological evidence on the air pollution health burden for the country, scored against a rubric rather than a single value.
- **T4 — Policy instruments.** Identification of policy programs, executing agencies, and legal instruments for air pollution control in force in the country.
- **T5 — Applied recommendation.** Rubric-scored applied recommendation for a defined operational scenario (for example, short-term response to a high-concentration episode under thermal inversion).

T1, T2, and T4 admit verifiable ground truth and carry the primary accuracy signal; T3 and T5 are rubric-scored open-generation tasks. The country-by-task mapping of official source classes, equivalence criteria, and verifiability status is specified in the ground-truth registry (Section 3.8).

3.6 Persona manipulation

Persona is a within-prompt factor with two levels. The **neutral** condition presents the query as a plain information request. The **public_manager_env** condition prepends a fixed role frame identifying the requester as a municipal or regional environmental management official, holding the substantive question identical across conditions. A single role-frame template is applied uniformly across countries and tasks (translated where the prompt language is non-English), so that any between-condition difference is attributable to the persona frame rather than to question wording. Persona order within each prompt-model cell is randomized, and a manipulation check verifies that the role frame is acknowledged in the response. The exact wording, translation procedure, and manipulation-check coding are pre-specified and reproduced in Supplementary Material. Persona is the basis of hypothesis H6, which tests whether the role frame reduces (or amplifies) the country-level accuracy gradient.

3.7 Prompt construction and validation

Prompts were authored for the air pollution policy domain by the research team with domain supervision (PPGSAU/UTFPR) and validated against the ground-truth registry before entering the confirmatory stimulus set. A pilot subset was first validated for inter-rater agreement using Krippendorff's α, with a threshold of $\alpha \geq 0.70$ required before scaling the design. All prompts were authored in English and, for a stratified subset of countries, additionally in a relevant native language (sparse multilingual matrix; Supplementary Table S2); native-language prompts were translated and back-translated by native-speaker translators, and any prompt whose back-translation produced a semantic shift was rewritten or excluded. Following the Brazil pilot validation, every factual claim in a ground-truth entry was required to resolve to an official source URL or a peer-reviewed DOI before the corresponding prompt was admitted, enforcing the project Citation Verification Protocol at the ground-truth generation stage rather than only at the manuscript stage.

3.8 Ground truth and the ground-truth registry

Ground truth for the verifiable tasks (T1, T2, T4) was drawn from official sources of record: national air quality standards and the agencies that issue them (T1), environmental-agency monitoring reports (T2), and federal policy programs with their executing agencies and legal instruments (T4). For T3, the reference is a curated set of peer-reviewed epidemiological studies with resolvable DOIs rather than a single number. Because the validity of the geographic gradient (H1) and the persona effect (H6) depends on ground truth being equally available and equally verifiable across all 15 countries, we constructed a per-country ground-truth registry that, for each (country, task) pair, records the class of equivalent official source, the resolvable URL, the gold value or rubric key, the reference date, the human validator, and a validation flag. The registry design, cross-country equivalence criteria, and the explicit separation of an AI knowledge gap from a ground-truth-availability gap are documented in `docs/ground_truth_registry_design.md` and summarized in Supplementary Material. Entries are versioned by access date, source URL, and a SHA-256 hash of the retrieved document. Where ground truth could plausibly have appeared in training data, a sensitivity subset is constructed from documents dated after each model's training cutoff. Entries whose official source is unavailable or non-equivalent for a country are flagged and excluded from the primary accuracy comparison for the affected (country, task) cell, so that missing ground truth is never scored as a model error.

3.9 Data collection

API and local calls were executed at temperature 0.3 with deterministic seeds where available, with 2 replications per prompt-model-persona combination to estimate stochastic variability within the budget. Collection occurred within a fixed window to minimize model-update confounds, with pinned tags and version strings recorded per call. All responses were stored alongside complete request metadata (model version string or tag, timestamp, token counts, finish reason, persona condition) in a versioned storage layer with SHA-256 checksums. Response quality gates excluded truncated responses (`finish_reason` $\neq$ `stop`) and responses in a language divergent from the request; both were retained and analyzed separately as secondary outcomes.

3.10 Measures

Scoring used a three-judge LLM-as-judge ensemble with a human gold layer. Each response was independently scored by three judge models against the ground-truth registry entry (for T1, T2, T4) or the validated rubric (for T3, T5); the ensemble score is the majority or mean across judges, and inter-judge agreement is reported via Krippendorff's α (target $\alpha \geq 0.70$). A blinded human gold subset, scored by domain-trained raters and adjudicated by a senior rater on disagreement, calibrates and audits the ensemble; judge-human agreement is reported as a validity check, and judges are not treated as authoritative where they diverge systematically from the human gold layer.

The primary composite outcome combines the five pre-specified subcomponents (see Supplementary Table S3), each normalized to $[0, 1]$: (i) factual accuracy against the registry entry (binary T1; partial credit T2–T4); (ii) contextual completeness scored against a pre-validated rubric (T3, T4, 0–5); (iii) citation quality verified via DOI and official-source check; (iv) hallucination absence, coded binarily (0 hallucinated, 1 faithful); and (v) applied validity, the rubric score for the operational recommendation in T5. Primary inference uses equal weights; two pre-specified sensitivity weightings (a PCA-derived set estimated on the BRA extended pilot and an author-specified set: factual 0.30, contextual 0.25, citation 0.15, hallucination 0.15, applied 0.15) are reported, and a conclusion is treated as robust only if it holds across all three. The composite is computed per (country, model, persona) cell to support the H6 difference-in-differences test.

3.11 Analytical strategy

The three primary confirmatory tests are pre-specified as a single family (F1) and corrected together with the Bonferroni-Holm procedure. Secondary confirmatory tests (F2) are reported unadjusted within family, and declared exploratory tests (F3) use Benjamini-Hochberg FDR at $q = 0.10$ [30].

H1 (geographic gradient, primary). Country-level mean composite accuracy (n=15) is regressed on the Joshi/HDI gradient via Spearman ρ, pre-specified one-sided with threshold $\rho \geq 0.55$ at $\alpha = 0.05$, with a Mann-Kendall trend test as a mandatory non-monotonicity complement and a Global South dummy as a sensitivity contrast.

H4 (corpus-representation mechanism, primary). Country-level partial Spearman ρ relates corpus representation to accuracy after adjusting for HDI and log GDP per capita, using Wikipedia counts as the primary proxy ($\rho \geq 0.55$) and requiring convergence across at least two of three Common Crawl operationalizations ($\rho \geq 0.40$). An E-value sensitivity analysis [14] quantifies the unmeasured confounding required to nullify the association (threshold 2.0).

H6 (persona effect, primary). The persona effect is tested as a country-level difference-in-differences on the composite outcome: $\Delta_{\text{GS}} = \overline{\text{persona}}_{\text{GS}} - \overline{\text{neutral}}_{\text{GS}}$ and $\Delta_{\text{GN}} = \overline{\text{persona}}_{\text{GN}} - \overline{\text{neutral}}_{\text{GN}}$, with H6 statistic $\Delta_{\text{GS}} - \Delta_{\text{GN}}$. H6 is supported (one-sided, $\alpha = 0.05$) if the persona condition reduces the Global South gap by at least 0.05 on the $[0, 1]$ composite (equivalently, 5 percentage points of absolute accuracy). Inference uses a country-level permutation test (5,000 permutations) reported alongside a parametric paired test; the two-sided test is reported as a secondary check because persona amplification is theoretically possible and is reported transparently if observed.

H3 (regional model, secondary). A pre-specified one-sided contrast in a generalized linear mixed model compares Cabra-Mistral 7B v3 on Brazilian Portuguese prompts against a scale-matched globally trained open model, with an exploratory companion contrast on other Lusophone contexts to distinguish gap closure from gap displacement.

H2 and H5 (exploratory). The prompt-language $\times$ country interaction (H2) and the open-frontier versus closed-frontier contrast (H5) are reported descriptively, with formal tests under F3 correction where applied.

Cross-cutting mixed-model analyses use Generalized Linear Mixed Models with random intercepts for country and model, implemented via `pymer4` [31] interfacing R's `lme4` [32]: `lmer` with REML for the Gaussian composite and `glmer` with a binomial family and logit link for binary outcomes. Where a mediation decomposition is reported for H4, it uses `semopy` [33] with 5,000 bootstrap replicates. Robustness analyses, all pre-specified, include leave-one-country-out re-estimation, restriction to the human-gold-validated subset, Bayesian re-estimation with weakly informative priors via `pymc` [34], and a separate analysis excluding prompts flagged for potential training-data contamination. A Monte Carlo power simulation (10,000 iterations under pilot-informed effect sizes) confirms family-wise power for the primary trio under Bonferroni-Holm; the within-subject persona design yields a minimum detectable gap reduction of approximately 2–3 percentage points, so the 5-point H6 threshold is well-powered. Power details are reported in Supplementary Material.

3.12 Study protocol and open science

The full protocol — hypotheses, prompts with hashes, the ground-truth registry, the persona template, scoring rubrics, GLMM and DiD specifications, multiple-comparison corrections, and exclusion criteria — is pre-specified in advance of confirmatory data collection. The analytic dataset, after quality gates but before model fitting, is frozen and hashed so that the confirmatory analysis can be audited against the locked specification. The full protocol, prompts, ground-truth references, analysis code, and anonymized response data will be released on publication under CC-BY-4.0 (data and prompts) and the MIT License (code) [35]; a companion Data Paper is prepared for *Scientific Data*.

3.13 Ethical considerations

No personal data from end-users of LLMs were collected. The study analyzes model outputs to publicly relevant policy questions; domain experts who contribute to ground-truth validation are acknowledged as co-authors when they meet ICMJE criteria. LLM provider terms of service were reviewed and respected, and responses are redistributed solely for research under fair-use and reproducibility provisions, excluded from any commercial application.

4 Results

Status note. This is a *15-country, 6-model, 4-language* benchmark. The core design crosses 15 countries, 6 models spanning the five access tiers (GPT-5, DeepSeek-V3, Claude Haiku, Qwen3-32B, Llama-3.1-8B, Cabra-Mistral-7B), 5 tasks, 2 persona conditions, and 4 languages (English plus a country-native language — Spanish, Portuguese, or Hindi — for the five applicable countries, giving the within-country English/native pairs that test H2). An *extended English-only arm* adds 7 further models (13 in total) for coverage robustness. All responses are scored by GPT-5-mini with an out-of-sample inter-judge reliability check; the corpus T1 ground truth is anchored to official primary sources for all 15 countries (Section 4.8). Two pre-specified items remain open and bound interpretation: the corpus-mechanism test (H4) is not analysed here, and the single judge is not yet the planned multi-judge ensemble plus human gold layer. Estimates are reported with effect sizes.

4.1 Sample and judging

The confirmatory set comprises the 150 air-pollution-policy prompts (15 countries $\times$ 5 tasks $\times$ 2 personas) with two replications per model. After excluding empty responses, the analysed corpus is 4,633 judge-scored responses on the $[0, 1]$ composite. The primary judge is GPT-5-mini. Because GPT-5-mini is itself among the evaluated tier, an out-of-sample judge (Claude Sonnet 4.6, not in the 14-model set) re-scored a stratified sample of 750 responses; agreement is strong (Spearman $\rho = 0.87$, ICC(2,1) $= 0.82$, Pearson $r = 0.86$, mean absolute difference 0.11), which neutralizes the judge–target overlap for the reported composites (Section 4.9).

4.2 Accuracy by model

Table 1 orders the 13 models by mean composite. Frontier systems (GPT-5, DeepSeek-V3, GPT-5-mini) lead; small open-weight models trail. The result bearing on H3 replicates the pilot and the Brazil collection: the Brazilian-Portuguese regional model Cabra-Mistral 7B is the *weakest* of all 13 (0.308), contradicting the optimistic H3a framing that a regional model buys domain accuracy. Scale and frontier training dominate regional instruction tuning at the 7B level.

Table 1: Confirmatory composite accuracy by model across 15 countries (GPT-5-mini judge, $[0,1]$). N = scored responses.

Model	N	Mean
GPT-5	580	0.641
DeepSeek-V3	300	0.622
GPT-5-mini	578	0.621
Claude Haiku 4.5	290	0.568
Qwen3 32B	300	0.523
GPT-OSS 120B	237	0.485
Llama 3.3 70B	274	0.482
Command R+	300	0.481
Phi-4 14B	300	0.460
Llama 4 Scout	300	0.449
Qwen3 14B	300	0.446
Llama 3.1 8B	577	0.409
Cabra-Mistral 7B	297	0.308

4.3 Accuracy by task: the factual-recall floor

Difficulty is sharply task-dependent (Table 2). The two hardest tasks are **T1 (technical-standard recall, 0.319)** and **T2 (local measured datum, 0.316)** — precisely the tasks that require retrieving a specific binding value (a national annual $PM_{2.5}$ standard; a city-year concentration). The open recommendation task (T5) scores highest (0.805) because its rubric rewards plausible, well-structured policy advice that does not hinge on a single fact. The pattern is the core empirical finding: LLMs are weakest exactly where applied environmental management needs them most — recalling the precise regulatory threshold or measured value in force.

Table 2: Confirmatory composite accuracy by task type.

Task	N	Mean
T5 (applied recommendation)	922	0.805
T3 (health-evidence synth.)	933	0.577
T4 (policy instruments)	906	0.540
T1 (technical standard)	935	0.319
T2 (local factual datum)	937	0.316

4.4 H1 — the geographic gradient

Aggregating the twelve Global South countries against the three Global North countries gives a Global South mean of **0.497** (n = 3,692) versus a Global North mean of **0.562** (n = 941): a gap of $+$**6.5 percentage points** (Global North minus Global South), Cohen's $d = +0.25$. The direction is consistent with H1 and replicates the pilot signal ($+6.0$ pp, $d = 0.34$) at full 15-country scale, now with the persona factor active. The effect is small. The country-level ordering (Table 3) is also informative and non-deterministic: India (0.632) tops all countries and Indonesia (0.539, both Global South) sit above Germany (0.531, Global North), while Argentina and Brazil anchor the bottom — partly because their ground truth is the most demanding (Argentina lacks a single binding federal standard; Brazil's standard moved to CONAMA 506/2024). The gradient is real in aggregate but not a clean North-over-South ranking.

Table 3: Confirmatory composite accuracy by country (tier: GN = Global North, GS = Global South).

Country	Tier	Mean	Country	Tier	Mean
India	GS	0.632	Peru	GS	0.490
Japan	GN	0.584	Kenya	GS	0.480
United States	GN	0.574	Egypt	GS	0.461
Indonesia	GS	0.539	Nigeria	GS	0.459
Germany	GN	0.531	Brazil	GS	0.443
South Africa	GS	0.512	Argentina	GS	0.441
Bangladesh	GS	0.501			
Philippines	GS	0.496			
Mexico	GS	0.492			

4.5 H6 — persona does not narrow the geographic gap

The persona factor was fully crossed in this collection, so H6 can be tested as a difference-in-differences. The public-environmental-manager frame moves Global South accuracy by $\Delta_{\mathrm{GS}} = -0.1$ pp and Global North accuracy by $\Delta_{\mathrm{GN}} = -0.8$ pp, giving a DiD of $\Delta_{\mathrm{GS}} - \Delta_{\mathrm{GN}} = \mathbf{+0.6}$ **pp** — far below the pre-specified 5 pp threshold for support. The geographic gap is essentially unchanged by persona: +6.8 pp under the neutral frame versus +6.2 pp under the manager frame. **H6 is not supported**: presenting the query as coming from a public environmental manager neither narrows nor amplifies the Global South disadvantage. This is consistent with evidence that expert personas do not reliably improve factual accuracy [4, 5], and it answers the study's central applied question in the negative — prompt framing is not a fix for geographic bias.

4.6 H5 — open versus closed-accessible

Grouping by access type, closed-accessible models average 0.604 ($n = 868$) against 0.461 for open-weight models ($n = 3{,}185$), a +14.3 pp closed advantage. As in the pilot, this runs against H5; here the open arm includes larger Tier A models (Llama 3.3 70B, DeepSeek-V3), so the gap is less attributable to scale alone, although DeepSeek-V3 itself ranks second overall. H5 is interpreted descriptively.

4.7 H2 — native-language prompting lowers accuracy

The 4-language arm renders each prompt for the five countries with a target native language (Brazil → Portuguese, Mexico/Argentina/Peru → Spanish, India → Hindi) in both English and the native language, holding content fixed, so the English/native contrast is within-country and within-model. Across 300 matched (model, prompt) pairs, native-language prompting *lowers* mean composite accuracy by 3.5 percentage points (English 0.506 → native 0.471). The effect is strongly modulated by the resource level of the native language (Table 4): for the high-resource Latin-script languages the penalty is small (Spanish −1.8 pp, Portuguese −0.5 pp), but for Hindi it is large (−11.6 pp). India, which scores highest of all countries in English (0.646), drops to 0.529 in Hindi. The applied implication mirrors H6: the intuitive fix — letting a local manager ask in their own language — does not help and, for a non-Latin-script lower-resource language, materially hurts. This is consistent with the linguistic-resource hierarchy [2] and motivates the corpus-representation mechanism (H4).

Table 4: H2: native-minus-English accuracy by native language and country (paired within country and model).

Language (Joshi)	Countries	Δ (pp)	N pairs
Spanish (5)	MEX, ARG, PER	−1.8	180
Portuguese (4)	BRA	−0.5	60
Hindi (4)	IND	−11.6	60
Overall		−3.5	300

4.8 Ground-truth provenance

Because measured "accuracy" depends on the ground truth, the technical-standard target (T1) for all 15 countries was anchored to official primary sources by documentary verification rather than expert assertion, and is reproducible from the cited URLs. Eleven countries have a value read from the official regulation (e.g. Brazil's annual $PM_{2.5}$ standard of 17 μg/m^3 read verbatim from Annex I of CONAMA Resolution 506/2024 in the Diário Oficial da União). Two countries have *no* national $PM_{2.5}$ standard, confirmed from the official text — Nigeria's NESREA regulation sets PM10 only, and Argentina has no binding federal value — a fact the audit established against a circulating but incorrect secondary figure. Two (Bangladesh, Egypt) have the governing law identified with the numeric value partially confirmed. This provenance is more reproducible than expert validation: any reader can re-verify each value from the official source.

4.9 Inter-judge reliability

To address the judge–target overlap (GPT-5-mini is in the evaluated set), an out-of-sample judge (Claude Sonnet 4.6) re-scored a stratified random sample of 750 responses spanning all countries, tasks, and personas. Agreement on the composite is strong: Pearson $r = 0.861$, Spearman $\rho = 0.874$, ICC(2,1) absolute agreement = 0.824, mean absolute difference 0.112. An ICC above 0.80 indicates good-to-excellent reliability and supports treating the GPT-5-mini composites as judge-independent for the effects reported here. The final confirmatory analysis will replace the single judge with the pre-specified multi-judge ensemble plus a human gold layer.

4.10 Summary

Five findings hold across the benchmark. (1) A small but consistent geographic gradient (+6.5 pp Global North advantage, $d = 0.25$), replicating the pilot. (2) Native-language prompting lowers accuracy (−3.5 pp overall, −11.6 pp for Hindi), so neither of the two intuitive remedies helps: persona is neutral and the native language hurts. (3) Persona prompting does not narrow the geographic gap (DiD +0.6 pp), answering the applied question negatively. (4) The regional Brazilian model is the weakest of all 13, contradicting H3a. (5) The failure concentrates in technical-standard and local-datum recall (T1, T2 $\approx$ 0.32), the regulatory-cutoff failure mode. Inter-judge reliability is strong (ICC 0.82), and the T1 ground truth is anchored to official primary sources for all 15 countries. These signals are robust to the judge but remain bounded by the open items in the status note — expert-validated ground truth for all 15 countries, the corpus-mechanism test (H4), the multilingual matrix (H2), and the pre-specified formal tests with multiplicity control — which are the work of the final confirmatory analysis. The geographic asymmetry on air pollution policy carries direct relevance for environmental management in the Global South, where the disease burden is large [1, 36] and corpus under-representation [2, 3] plausibly shapes what these models know.

5 Discussion

This paper contributes a study protocol and a pilot. The protocol pre-specifies a benchmark for air pollution policy with verifiable official ground truth, a persona manipulation as a within-subject experimental factor, and a planned confirmatory study across 15 theoretically stratified countries. The pilot exercises that protocol end-to-end on a reduced configuration (7 countries, 5 models, neutral prompts only, single automated judge) to validate the pipeline and to produce a first directional signal. We read the pilot as evidence about *feasibility and plausible direction*, not as a test of any hypothesis. The numbers below are exploratory, the inference is noisy, and the confirmatory study is what will decide the questions we pose. We organise the discussion around the pilot signal, the persona hypothesis and why it is worth testing, the public-health stakes that motivate the domain, the limitations that bound what the pilot can say, and the confirmatory study these results are designed to inform.

5.1 Reading the pilot signal

The geographic gap (H1) appears, in the expected direction

In the pilot, mean composite accuracy was higher for the two Global North countries (0.636, $n = 200$) than for the five Global South countries (0.575, $n = 500$), an absolute gap of about 6 percentage points (pilot Cohen's $d \approx 0.34$, pooled SD 0.180). The direction matches H1 and is consistent with the broader literature on geographic factual bias: WorldBench's elevated error ratio for Sub-Saharan Africa versus North America [8] and Global-Liar's privileging of Global North statements [10]. That the pattern surfaces in an applied air pollution policy task suite, rather than in numeric trivia or journalistic fact-checking, is the feasibility result we sought: the tasks discriminate, the ground truth anchors, and the pipeline returns a signal pointed the way the theory predicts.

We do not over-read the magnitude. Six points is small-to-moderate, the between-country variance in the pilot was modest ($\text{ICC}_{\text{country}} \approx 0.03$), and with only seven countries and a single judge the country-level estimates are noisy. A 6-point pilot gap is compatible with a range of confirmatory outcomes, including a null. The pilot's role is to show that the effect is large enough to be worth powering for, and the observed variance components feed directly into the confirmatory power analysis.

The open-versus-closed contrast (H5) ran against expectation

The pilot's H5 signal points the other way from what the protocol entertained. Closed accessible models (Claude Haiku, GPT-5-mini, Gemini Flash; mean 0.643) outscored open frontier models (Llama 4 Scout, Command R+; mean 0.517) by about 12.5 percentage points, well beyond the 5-point band the protocol treats as practical equivalence. We flag this as a contrary signal and resist explaining it away, but two features of the pilot configuration bear directly on it. The open arm was represented by Llama 4 Scout at 17B parameters, smaller than the 70B-and-above frontier open models the confirmatory study targets, so the contrast partly reflects model scale rather than the open-versus-closed property of interest. H5 is also an exploratory hypothesis in the protocol, reported descriptively rather than as a powered confirmatory test. The honest reading is that the pilot does not support a parity claim for open models at this scale, and the confirmatory study, with larger open models in the arm, will provide the cleaner contrast.

Task-level behaviour is informative for the protocol

Composite accuracy varied substantially across the five task types in the pilot, with the normative-recall task (T1, mean 0.507) and the applied recommendation task (T5, mean 0.499) hardest, and the health-evidence synthesis task (T3, mean 0.677) easiest. T1 anchors to the binding ambient standard, which in Brazil sits inside an active regulatory transition: CONAMA Resolution 506/2024 supersedes the 491/2018 framework and brings the interim targets into force on a defined schedule. A model that has absorbed the older framework but not the transition will recall a superseded value, which is precisely the kind of high-stakes, time-sensitive error a normative task is built to surface. The low T1 score is consistent with that mechanism, though the pilot cannot attribute it, and it confirms that the normative task discriminates as intended.

5.2 Why test persona (H6)

The protocol's co-primary hypothesis, H6, asks whether adopting the persona of a public environmental manager changes the Global South accuracy gap. The pilot did not manipulate persona; all pilot prompts were neutral. H6 is therefore a question the confirmatory study will answer, and the case for asking it rests on a tension in the evidence.

Practitioners prefix queries with roles routinely ("As a municipal environmental secretary, ..."), on the intuition that a role frame steers the model toward policy-relevant, norm-compliant answers. The accuracy literature does not support that intuition. Across four model families and 2,410 factual questions, adding expert personas to system prompts did not improve accuracy over a no-persona control, and averaged across 162 roles it slightly reduced it [4]; an independent analysis reaches the same conclusion that expert personas do not improve factual accuracy [5]. If that result holds in our domain, the naive expectation that a persona "helps" is wrong, and the open question becomes which way it fails.

We pre-specify both directions. Under the *Norm-Compliance* hypothesis, the manager persona narrows the Global South gap by cueing the model toward the regulatory register where official standards and agencies live. Under the *Persona-Vacuum* hypothesis, the persona widens the gap by inviting confident fabrication of regulatory specifics the model never learned for under-represented countries, converting an honest "I don't know" into a fluent but wrong attainment deadline or agency name. The persona literature gives us no reason to expect a free accuracy gain and a concrete reason to worry about the second failure mode, which is exactly why persona belongs in the design as a controlled factor rather than an uncontrolled habit of practice. Treating it as an experimental lever lets the confirmatory study report whichever direction the data support, with the amplification direction reported transparently if it appears.

5.3 Why the domain matters

The stakes are not benchmarking abstractions. Fine particulate matter is among the leading environmental risk factors for premature mortality, and the burden falls disproportionately on the Global South; the Global Burden of Disease estimates attribute on the order of fifty thousand deaths per year in Brazil alone to $PM_{2.5}$ exposure [1], and short-term exposure continues to drive measurable mortality and morbidity in Brazilian cities [36]. When a public environmental manager in a Global South city delegates a normative lookup, a measured-concentration retrieval, or an operational recommendation to an LLM, an inaccurate answer can propagate into a regulatory brief or an emergency response during a pollution episode. A geographic gap of even a few points in this setting is consequential in a way it is not for cultural trivia, because the under-served languages and regions that sit in the lower classes of the Joshi resource hierarchy [2] are also where air pollution kills the most people. That alignment between corpus under-representation and mortality burden is the link the protocol is built to probe, and it is what makes the confirmatory study worth running carefully rather than quickly.

5.4 Limitations

The pilot's limitations are substantial and define the boundary between what we observed and what we can claim.

Single automated judge. Pilot scoring used one LLM judge (Claude Haiku 4.5) under a G-Eval rubric. A single judge yields no inter-rater reliability estimate, and the judge model is itself in the evaluated sample, raising a self-evaluation concern that the pilot documents but cannot resolve. The confirmatory study uses a non-overlapping judge and a second rater (or human panel) with a pre-specified agreement threshold.

No persona condition. The pilot ran neutral prompts only. The co-primary persona hypothesis (H6) is therefore untested here; the pilot speaks to feasibility and to H1/H5 direction, not to persona effects.

No multilingual prompts. All pilot prompts were in English. The language $\times$ geography interaction (H2), which is where the resource hierarchy is expected to bite hardest, is not exercised in the pilot.

Seven countries, single-author non-Brazilian prompts. The pilot covers 7 of the 15 protocol countries. Prompts for non-Brazilian countries were authored by a single author without the regional expert-panel validation the protocol requires, so cross-country comparisons are provisional and the country-level estimates are noisy at this N.

Reduced open-model arm. The open arm used Llama 4 Scout at 17B, smaller than the frontier open models (70B and above) the protocol specifies. The H5 contrast is consequently confounded with scale and should not be read as a verdict on open models in general.

These are limitations of the pilot, not of the protocol. Each is addressed by the confirmatory configuration: multi-rater scoring with a non-overlapping judge, the full persona manipulation, a sparse multilingual matrix, expert-panel prompt validation across all 15 countries, and frontier-scale open models in the open arm.

5.5 Toward the confirmatory study

The pilot does its job: it shows the pipeline runs end-to-end, the air pollution policy tasks discriminate, the H1 gap appears in the expected direction at a magnitude worth powering for, and the variance components needed to finalise the power analysis are now in hand. It also surfaces a contrary H5 signal that the confirmatory design, with larger open models, is positioned to test cleanly. The planned confirmatory study executes the pre-specified protocol in full: 15 countries stratified on UNCTAD North/South classification, the Joshi linguistic-resource taxonomy [2], and development indicators; the persona manipulation as a within-subject factor for the co-primary H6 test; a non-overlapping judge with inter-rater reliability; and the corpus-representation mechanism test with E-value sensitivity. We interpret the whole programme within the coloniality-of-knowledge framework [3]: LLMs are compressed summaries of corpora produced under uneven digital and publication access, and the confirmatory study tests one link in a theorised reproduction cycle, from training-corpus asymmetry to LLM knowledge asymmetry to policy decision asymmetry, while the persona manipulation tests whether prompt framing interrupts or aggravates that link. Until the confirmatory data are collected, the appropriate conclusion is a methodological one: the design is feasible, the direction is plausible, and the questions are worth the larger study.

6 Conclusion

Large language models are becoming working infrastructure for air pollution policy, a domain where an inaccurate answer about a binding standard, a measured concentration, or an operational response carries direct public-health stakes in the Global South, where the $PM_{2.5}$ mortality burden is heaviest [1, 36]. This paper contributes a pre-specified study protocol for measuring geographic and persona-modulated bias in that domain, together with a pilot that exercises the protocol end-to-end.

The pilot is exploratory and small: 7 countries, 5 models, neutral prompts, and a single automated judge. Within those bounds it is encouraging for the protocol. The pipeline runs end-to-end, the air pollution policy tasks discriminate across countries and task types, and the Global South accuracy gap appears in the expected direction at about 6 percentage points, a magnitude worth powering a confirmatory study for. The pilot also returns a contrary signal on the open-versus-closed model contrast, plausibly because the open arm used a model smaller than the confirmatory target, which the larger study is designed to test cleanly.

We do not claim a confirmed result. The pilot did not manipulate persona, used English-only prompts, relied on a single judge that overlaps the evaluated models, and drew on single-author prompts for non-Brazilian countries. The persona question (H6) is open, and the persona literature gives us reason to expect no free accuracy gain and a concrete risk of amplified fabrication for under-represented countries [4, 5], which is why we test both the norm-compliance and persona-vacuum directions in the confirmatory design rather than assuming a persona helps.

The contribution here is a defensible foundation: a domain benchmark with verifiable official ground truth, a persona manipulation specified as a controlled experimental factor, theory-driven country sampling, and a pilot that demonstrates feasibility and a plausible direction. The confirmatory study will execute the full protocol across all 15 countries with persona, multilingual prompts, expert-validated stimuli, frontier-scale open models, and non-overlapping multi-rater scoring. We interpret the programme within the coloniality-of-knowledge framework [3]: the languages and regions under-served in training corpora [2] are the same ones where air pollution kills the most people, and whether prompt framing interrupts or aggravates that asymmetry is the question the larger study is built to answer. The protocol, pilot data, code, and prompts are released openly to support that study.

Data and code availability

The full protocol, prompts, and analysis code will be released on publication at github.com/Roverlucas/artigo-vies-analise-regional under MIT (code) and CC-BY-4.0 (data/prompts/documentation) licenses, together with the ground-truth references and (upon real execution) anonymised LLM responses. The analysis pipeline is reproducible end-to-end via `python code/run_all.py`.

Study protocol

The full statistical analysis plan is pre-specified in the study protocol. All confirmatory analyses follow the pre-specified specifications; the present paper reports only pilot evidence, and any exploratory analysis is labeled explicitly and corrected separately. The full protocol, prompts, and analysis code will be released on publication.

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Author contributions

Following CRediT taxonomy:

- **Lucas Rover (UTFPR):** Conceptualization, Methodology, Software, Data Curation, Formal Analysis, Investigation, Visualization, Writing – Original Draft, Project Administration.
- **Dr. Eduardo Tadeu Bacalhau (UFPR):** Validation, Supervision, Writing – Review & Editing.

- **Dra. Yara de Souza Tadano (UTFPR):** Validation, Supervision, Writing – Review & Editing, Funding Acquisition.

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Declaration of interests

The authors declare no competing interests.

References

[1] GBD 2019 Risk Factors Collaborators. Global burden of 87 risk factors in 204 countries and territories, 1990–2019: a systematic analysis for the Global Burden of Disease Study 2019. *The Lancet*, 396(10258):1223–1249, 2020. doi: 10.1016/S0140-6736(20)30752-2.

[2] Pratik Joshi, Sebastin Santy, Amar Budhiraja, Kalika Bali, and Monojit Choudhury. The state and fate of linguistic diversity and inclusion in the NLP world. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 6282–6293. Association for Computational Linguistics, 2020. doi: 10.18653/v1/2020.acl-main.560.

[3] Shakir Mohamed, Marie-Therese Png, and William Isaac. Decolonial AI: Decolonial theory as sociotechnical foresight in artificial intelligence. *Philosophy & Technology*, 33(4):659–684, 2020. doi: 10.1007/s13347-020-00405-8.

[4] Mingqian Zheng, Jiaxin Pei, Lajanugen Logeswaran, Moontae Lee, and David Jurgens. When "a helpful assistant" is not really helpful: Personas in system prompts do not improve performances of large language models. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 15126–15154, 2024. doi: 10.18653/v1/2024.findings-emnlp.888.

[5] Lennart Meincke, Basir Savir, Ina Shapiro, Dan Shapiro, Lilach Mollick, and Ethan R. Mollick. Prompting science report 4 — playing pretend: Expert personas don't improve factual accuracy. SSRN Working Paper, 2025. SSRN id 5879722.

[6] Lisa Messeri and M. J. Crockett. Artificial intelligence and illusions of understanding in scientific research. *Nature*, 627(8002):49–58, 2024.

[7] Christopher A. Bail. Can generative AI improve social science? *Proceedings of the National Academy of Sciences*, 121(21), 2024.

[8] Mazda Moayeri, Elham Tabassi, and Soheil Feizi. Worldbench: Quantifying geographic disparities in LLM factual recall. In *Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency*, pages 1211–1228, Rio de Janeiro, Brazil, 2024. doi: 10.1145/3630106.3658967.

[9] Rohin Manvi, Samar Khanna, Marshall Burke, David Lobell, and Stefano Ermon. Large language models are geographically biased. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, 2024. arXiv:2402.02680.

[10] Shujaat Mirza, Bruno Coelho, Yuyuan Cui, Christina Pöpper, and Damon McCoy. Global-liar: Factuality of LLMs over time and geographic regions. *arXiv preprint arXiv:2401.17839*, 2024.

[11] Emily M. Bender, Timnit Gebru, Angelina McMillan-Major, and Shmargaret Shmitchell. On the dangers of stochastic parrots: Can language models be too big? In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*, pages 610–623, 2021. doi: 10.1145/3442188.3445922.

[12] Junho Myung et al. BLEnD: A benchmark for LLMs on everyday knowledge in diverse cultures and languages. *Advances in Neural Information Processing Systems*, 37:78104–78146, 2024.

[13] Brian A. Nosek, Charles R. Ebersole, Alexander C. DeHaven, and David T. Mellor. The preregistration revolution. *Proceedings of the National Academy of Sciences*, 115(11):2600–2606, 2018. doi: 10.1073/pnas.1708274114.

[14] Tyler J. VanderWeele and Peng Ding. Sensitivity analysis in observational research: introducing the E-value. *Annals of Internal Medicine*, 167(4):268–274, 2017. doi: 10.7326/M16-2607.

[15] Aníbal Quijano. Coloniality of power, eurocentrism, and Latin America. *Nepantla: Views from South*, 1(3):533–580, 2000.

[16] Nick Couldry and Ulises A. Mejias. *The costs of connection: How data is colonizing human life and appropriating it for capitalism.* Stanford University Press, 2019.

[17] Catherine D'Ignazio and Lauren F. Klein. *Data feminism.* MIT Press, 2020.

[18] Unspecified Authors. BRoverbs: Measuring how much LLMs understand Portuguese proverbs. In *Journal of the Brazilian Computer Society*, 2025.

[19] T. Almeida et al. TiEBe: Temporal and Regional event benchmark for LLMs. *arXiv preprint*, 2025.

[20] Unspecified Authors. ALBA: A european portuguese benchmark for evaluating language and linguistic dimensions in generative LLMs. *arXiv preprint arXiv:2603.26516*, 2026.

[21] AMALIA team. AMALIA technical report: A fully open source large language model for european portuguese. Technical report, arXiv:2603.26511, 2026.

[22] T. Almeida, R. Nogueira, and H. Pedrini. Building high-quality datasets for Portuguese LLMs: From Common Crawl snapshots to industrial-grade corpora. *Journal of the Brazilian Computer Society*, 31(1):1247–1263, 2025.

[23] M. Opuszko and P. Böhm. New york, new york – unraveling bias in large language models: Investigating differences between standard and reasoning-based language models. In *AI Revolution: Research, Ethics and Society*, pages 92–106. Springer, 2026.

[24] F. Kerche, M. Zook, and M. Graham. The silicon gaze: A typology of biases and inequality in LLMs through the lens of place. *Platforms & Society*, 2026. doi: 10.1177/29768624251408919.

[25] Jiao He et al. What would an LLM do? evaluating policymaking capabilities of large language models. *arXiv preprint arXiv:2509.03827*, 2025.

[26] Z. Kozlakidis, T. Wootton, and M. T. Mayrhofer. Through the looking glass: ethical considerations regarding LLM-induced hallucinations to medical questions. *Frontiers in Digital Health*, 8:1736616, 2026.

[27] Michael Quinn Patton. *Qualitative research and evaluation methods*. SAGE Publications, 4th edition, 2015.

[28] UNCTAD. UNCTAD statistical classifications — june 2024 update. Technical report, UN Trade and Development, 2024. URL https://unctadstat.unctad.org/EN/Classifications.html.

[29] World Bank Group. World Bank Country and Lending Groups — FY26 Classification. Technical report, The World Bank, 2025. URL https://datahelpdesk.worldbank.org/knowledgebase/articles/906519.

[30] Yoav Benjamini and Yosef Hochberg. Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B*, 57 (1):289–300, 1995. doi: 10.1111/j.2517-6161.1995.tb02031.x.

[31] Eshin Jolly. pymer4: Connecting R and Python for linear mixed modeling, 2018.

[32] Douglas Bates, Martin Mächler, Ben Bolker, and Steve Walker. Fitting linear mixed-effects models using lme4. *Journal of Statistical Software*, 67(1):1–48, 2015. doi: 10.18637/jss.v067.i01.

[33] Anna A. Igolkina and Georgy Meshcheryakov. semopy: A python package for structural equation modeling. *Structural Equation Modeling*, 27(6):952–963, 2020.

[34] Oriol Abril-Pla, V. Andreani, C. Carroll, Lan Dong, C. J. Fonnesbeck, M. Kochurov, Ravin Kumar, J. Lao, C. C. Luhmann, O. A. Martin, et al. PyMC: A modern and comprehensive probabilistic programming framework in Python. *PeerJ Computer Science*, 9:e1516, 2023.

[35] Lucas Rover, Eduardo Tadeu Bacalhau, and Yara Tadano. Geographic bias in LLMs: Benchmark dataset and code, 2026. To be deposited on publication.

[36] Weeberb J. Requia, Ana Maria Vicedo-Cabrera, Heresh Amini, and Joel D. Schwartz. Short-term air pollution exposure and mortality in Brazil: Investigating the susceptible population groups. *Environmental Pollution*, 340:122797, 2024. doi: 10.1016/j.envpol.2023.122797.